```
import cv2
from google.colab.patches import cv2_imshow # Import cv2_imshow for Colab compatibility

# NOTE: Please upload your image file (e.g., image1.jpg) to the Colab environment.
# A common location is /content/. Make sure the path below matches your uploaded file.
img = cv2.imread('/content/Screenshot 2026-07-27 100608.png') # Updated image path

# Check if the image was loaded successfully
if img is None:
    print("Error: Image not found or could not be loaded. Please check the path and ensure the image exists.")
else:
    gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    cv2_imshow(gray_img) # Correctly using cv2_imshow without a title argument
    cv2.imwrite('gray_image.jpg', gray_img)
    # cv2.waitKey(0) and cv2.destroyAllWindows() are not needed with cv2_imshow in Colab
```

Apply Gaussian Blur

Gaussian blurring is a common technique to smooth images and reduce noise. It uses a Gaussian function to calculate the transformation to apply to each pixel in the image.

Gemini

```
# Apply Gaussian Blur to the grayscale image
# The kernel size (e.g., (5, 5)) determines the amount of blur.
# The third argument is the standard deviation in X and Y directions.
blurred_img = cv2.GaussianBlur(gray_img, (5, 5), 0)

cv2_imshow(blurred_img)
cv2.imwrite('blurred_image.jpg', blurred_img)

print("Blurred image displayed and saved as 'blurred_image.jpg'.")
```

```
Blurred image displayed and saved as 'blurred_image.jpg'.
```

Double-click (or enter) to edit